Zhaoyang Jia

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CS Ph.D. student at UC San Diego, specializing in machine learning, optimization, and turning genomics problems into tractable models, with novel methods validated on biobank-scale data and a record of interdisciplinary collaboration.

Education

University of California, San Diego

Ph.D. Computer Science and Engineering | GPA 4.00/4.00 Sep. 2023 – Dec. 2028 (expected)
Focus: Applied Machine Learning & Computational Genomics | Advisor: Vineet Bafna

B.S. Mathematics–Computer Science & B.S. Molecular and Cell Biology | GPA 3.95/4.00 Sep. 2019 – Aug. 2023

Skills

Machine Learning: PyTorch, Sequence Modeling (CNN/BiLSTM multi-class classifiers), Unsupervised Clustering
Algorithms & Optimization: Graph Theory, Integer Linear Programming, Dynamic Programming
Languages & Libraries: Python (Pandas, NumPy, Scikit-learn), Java, SQL; agentic coding workflows
HPC: SLURM, Linux, Bash, Git, AWS S3, Python multiprocessing, simulation-driven test harness

Publications and Conference Presentations

(*joint first author)

- **Z. Jia**, T. Löwenstern, ... V. Bafna. Accurate detection of Robertsonian translocations using ONT long-reads by estimating CN-loss at acrocentric satellite junctions. *(preprint expected Nov. 2026)*. **Lightning talk** and poster, ASHG Annual Meeting, Montreal, Oct. 2026.
- S. Raeisi Dehkordi*, **Z. Jia***, J. Estabrook, ... V. Bafna (2025). OMKar automates genome karyotyping using optical maps to identify constitutional abnormalities. *Genome Research*. doi:10.1101/gr.280536.125. Poster, American Cytogenomics Conference, Norfolk, June 2024.
- Z. Zhu, W. Han, Y. Deng, **Z. Jia**, ... T. M. Rana (2026). microRNA-25 drives immune checkpoint therapy resistance by repressing innate and humoral immunity via Syndecan-3. *Nature Communications*. doi:10.1038/s41467-026-73339-y.

Research Experience

Graduate Student Researcher – *ML & Algorithms for Genomics*, Bafna Lab, UC San Diego Jul. 2023 – Present

- Built a CNN+BiLSTM pre-filter that screens raw whole-genome sequencing for repetitive satellite-junction reads at ∼100% recall and keeps 1 in 20,000 reads, cutting runtime 25× compared to the standard mapping pipeline; applied it to screen 60 TB of biobank data (ASHG Annual Meeting, 2026);
- Implemented unsupervised k-medoids and hierarchical clustering to discover micro-signatures across duplicated regions in the human genome;
- Built a modular and scalable pipeline for detecting Robertsonian translocations from raw sequencing data, identifying all 8 carriers (100% sensitivity) with 98% specificity across 224 controls.
- Co-developed OMKar (*Genome Research*, 2025), an ILP-based (Integer Linear Programming) automated karyotyping and visualization tool that consolidates two sets of variant calls and recovers missed calls; correctly karyotyped 144 of 154 clinical samples and reached 95% SV recall and 88% precision across 250 simulated high-complexity clusters;
- Built a simulation-driven test harness for OMKar's automated regression testing across versions, generating up to three whole-genome simulations per minute with parameterized random rearrangements.

Undergraduate Researcher – *Bioinformatics Pipelines & Analyses*, Rana Lab, UC San Diego Jun. 2022 – Jul. 2023

- Built reproducible, parameterized pipelines for bulk, small-RNA, single-cell RNA-seq, and iCLIP data, automating the full path from raw data to analysis-ready count matrices and QC reports.
- Collaborated with wet-lab biologists from project conception through analysis, co-designing experiments and iterating on pipeline features as practical lab needs evolved (*Nature Communications*, 2026).

Undergraduate Researcher – *Wet-Lab Genome Editing*, Wu Lab, Scripps Research Jun. 2021 – Jul. 2022

- Designed and generated CRISPR knockouts in U2OS double-strand-break reporter cells to study break-induced replication (BIR), established validation assays, and ran NGS and Sanger sequencing on the reporter constructs.
- Built a sequencing analysis pipeline to detect abnormal repair products and quantify BIR template switching.

Projects

Multi-Agent Vehicle Trajectory Forecasting

- Designed a three-branch network for ego-vehicle trajectory forecasting: transformer attention over 50 surrounding agents, an LSTM over ego motion, and a CNN over the rasterized scene, fused by a differentiable bilinear lookup.
- Cut prediction error by 30% over the LSTM-only baseline with rotation/mirror augmentation, relative-motion features, and post-processing for a train/test shift in halted scenes.

Glioma Marker-Gene Prediction from TCGA Transcriptomes

- Designed a coarse-to-fine recursive feature elimination schedule with ridge regression to predict seven glioma marker genes from a 60,660-gene RNA-seq expression matrix (408 TCGA samples).
- Reduced held-out test MSE by 19–82% across all seven targets; SHAP attribution recovered known cell-cycle co-expression for MKI67, and survival analysis of 102 patients (regression and Kaplan–Meier) associated MIB1 and IDH1 expression with overall survival ($p = 0.01, 0.03$).

Leadership Experience

Project Mentor, Bafna Lab, supervised four bachelor/master students on modules of my project — Jun. 2024 – Present
Teaching Assistant, CSE 284: core course of the Bioinformatics Ph.D. program — Apr. 2024 – Jun. 2024
Mentor, Mentor Collective Program, UC San Diego, advised 12+ students on career and research — Sep. 2022 – Present